Online Exam for the course *Logic and Sets*
June 4, 2020, 18:30–22:00

(Students who only hand in solutions for part 2 because they want to reuse their mark for the midterm exam, have until 20:45.)

This exam has 4 pages and 8 exercises.

The result will be computed as (the total number of points plus 10) divided by 10.

Please motivate all answers!

By participating in this exam, I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to:

- make this exam completely on my own, and not share my solutions with other students,
- not consult sources other than explicitly mentioned in the examination, and
- make myself available for any oral explanation of my answers.

Part 1

1. Island puzzle (10 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A, B, and C.

A says: "Either B or C is a liar." (This is an exclusive or, i.e., $\oplus$.)

B says: "If C is a liar, then I am a truth speaker."

C says: "A is a truth speaker."

Determine, either through logical reasoning or via a truth table, who of A, B, and C is a liar and who is a truth speaker.

If you use logical reasoning, then at the end also check that the status of the islanders (liar or truth speaker) is in line with the truth values of their statements.

2. CNF and DPLL (4.5 + 8 points)

(a) Extract a formula ϕ in CNF from the following truth table:

p	q	ϕ
T	T	F
T	F	F
F	T	T
F	F	F

(b) Apply the DPLL procedure to this CNF to try to find a satisfaction of this formula. Why are you not surprised by the outcome?

3. Sets (5.5 + 5 points)

(a) Suppose that in a universe U with 28 elements, there are three sets A, B and C of which we know that

$$\#A = 10 \qquad \#B = 14 \qquad \#C = 15$$
$$\#(A \cap B) = 7 \qquad \#(B \cap C) = 11 \qquad \#(C \cap A) = 6$$
$$\#(A \cap B \cap C) = 6$$

Draw a Venn diagram of the sets A, B and C, determine the number of elements in every region of your Venn diagram, and use your Venn diagram to determine

$$\#(B \cup C) \qquad \text{and} \qquad \#(A \setminus (B \cap C))$$

(b) Using the rules of the algebra of sets, prove that the following set-theoretic equality holds (for all sets A, B, and C):

$$(B \setminus A)' \cup (C \setminus A)' = (B \cap C)' \cup A$$

4. Ordering relations (8 + 2 + 2 points)
Consider the ordering relation *IsDivisorOf* on the set A defined by

$$A := \{2, 3, 4, 5, 6, 10, 12, 15, 60\}.$$

(a) Use the algorithm that you have learned to construct the Hasse diagram that represents the *IsDivisorOf* ordering relation on A: explicitly write down the sets G_x and H_x (with $x \in A$) obtained in the construction, and then draw the Hasse diagram.

(b) Does the set A have a smallest element according to the ordering relation? If so, give this smallest element. If not, list all minimal elements of A.

(c) Does the set A have a largest element according to the ordering relation? If so, give this largest element. If not, list all maximal elements of A.

Part 2

5. OBDDs (2 + 4.5 + 6 points)
Let the propositional formula ϕ, with propositional variables x, y, z, have the following truth table:

x	y	z	ϕ
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	F
F	F	F	T

(a) Give the binary decision tree for ϕ, against the variable order x, y, z.

(b) Reduce this binary decision tree to a reduced OBDD. Clearly specify which reduction steps you apply.

(c) Construct an OBDD for $\forall x\, \phi$ from the reduced OBDD for ϕ, and reduce the resulting OBDD.

6. Predicate logic (5 + 5 points)
For each of the following pairs of formulas, either argue that they are semantically equivalent, or give a model on which their truth values are different. In case semantic equivalence does not hold, explain moreover whether one of the formulas semantically entails the other.

(a) $\neg(\exists x\, C(x) \vee \exists x\, D(x))$ and $\forall x\, (\neg C(x) \wedge \neg D(x))$

(b) $\neg(\exists x\, C(x) \wedge \exists x\, D(x))$ and $\forall x\, (\neg C(x) \vee \neg D(x))$

7. Functions (3.5 + 4 + 4 points)
We are given the functions $inv\colon \mathbb{R} \to \mathbb{R}$, $dec\colon \mathbb{R} \to \mathbb{R}$, and $exp_2\colon \mathbb{R} \to \mathbb{R}$ defined by

$$inv(x) := \frac{1}{x} \qquad dec(x) := x - 1 \qquad exp_2(x) := 2^x$$

(a) Consider the function f defined by $f := inv \circ inv$. What is the domain of definition D_f of f, and what is its range R_f? Please also give an explicit formula for $f(x)$ (valid for all values of x in the domain of definition).

(b) Give an explicit formula for $g(x)$, where g is the function defined by

$$g := inv \circ dec^{-1} \circ exp_2.$$

(c) Express the function $h\colon \mathbb{R} \to \mathbb{R}$ defined by

$$h(x) := 2^{\frac{1}{x}} - 2$$

as a composition of the functions inv, dec, exp_2 and their inverses.

8. Induction (3 + 8 points)
Consider the sequence $(t_n)_{n=1}^{\infty}$ of integer numbers defined recursively by

$$t_1 := 2, \qquad t_{n+1} := 2 \cdot t_n + 2 - n.$$

We claim that for all $n \geq 1$,

$$t_n = 2^n + n - 1.$$

(a) Verify by explicit calculation that the claim is true for $n = 1$, $n = 2$ and $n = 3$.

(b) Prove by mathematical induction that the claim is true for *all* integers $n \geq 1$.